

AMIRIS Q&A

Direct, detailed answers to clarification questions raised while reviewing the Progress Report and supervisor deck

9 September 2026

Every answer below was checked directly against the real code, agent configuration files, or result data at the time of writing - not answered from memory alone. Where a figure or formula is stated, it was re-confirmed (re-reading the actual file, or re-running the actual analysis script) rather than assumed to still be accurate.

1. How the Import Model Was Fixed and Calibrated

You asked: "How did we fix and calibrate the import model? Explain the code tweaked or process gone through."

The bug (Phase 7)

The original import availability was not a flat ceiling - it was shaped hour-by-hour from real 2023 cross-border flow TIMING. Checking a real shortage hour directly showed the import trader offering 0 MW at exactly the hour AMIRIS ran short. Checked systematically: 301 of 469 shortage hours (64%) had zero import available - because 2023's real flow pattern has no relationship to when this synthetic 2027 build happens to run short.

The fix (`build_2027_import_fix.py`)

Decoupled import availability from any historical timing entirely. `AvailableEnergyForImport.csv` became a flat constant (e.g. 30000.0000 every single hour), and `ImportCostInEURperMWh.csv` kept the real 2023 French day-ahead price series unchanged - so the CEILING is fixed, but how much of it actually gets used each hour is a genuine market outcome (AMIRIS's ImportTrader only imports when it is cheaper than domestic generation).

Calibration

Swept 20,000 / 25,000 / 30,000 / 37,650 MW ceilings against Brainpool's real price. Found V1 and V2 needed DIFFERENT ceilings - V2 (better demand model) settled on 30,000 MW; V1 (cruder demand, higher structural peak) needed the full 37,649.98 MW (the real observed peak single-hour German gross import in 2023). Picked by MAE/correlation, not bias alone, since bias can hide poor hour-to-hour tracking behind cancelling errors.

2. The Weekday/Weekend Mismatch

You asked: ""Found and fixed a weekday/weekend mismatch" - explain the mismatch and how it was fixed."

The mismatch

The granular breakdown (Phase 10) found weekday hours biased -22.78 EUR/MWh vs. Brainpool, weekend hours +16.18 - a ~39 EUR/MWh systematic gap, even though correlation WITHIN each group was strong (0.64+). That pattern means the demand shape's calendar was misaligned, not that the shape itself was wrong.

Root cause

The 2023 source data was mapped onto 2027 POSITIONALLY - hour 1 of 2023 became hour 1 of 2027, regardless of actual weekday. 2023-01-01 was a Sunday; 2027-01-01 is a Friday. So every 'weekday' in the model's calendar was actually built from a mix of real weekdays and weekends.

The fix (build_2027_demand_v2_weekday.py)

Shifted the 2023 source data by 5 days (120 hours, a circular roll) so 2023-01-06 (a Friday) becomes the new start - aligning every subsequent day's real weekday with the corresponding 2027 day. One tiny documented imperfection: the last ~5 days of the year wrap around a day off ($365 \bmod 7 = 1$), affecting under 0.1% of the year.

3. The Two Subsidy Types and Their Oversupply Behaviour

You asked: "Two solar/wind subsidy types behave very differently under oversupply" - what is the subsidy type and what was their behaviour?"

AMIRIS supports two real subsidy mechanisms, and this project uses both:

- * MPVAR (market premium, variable) - the plant bids into the market like anyone else, gets whatever the market clears at, and receives a top-up subsidy on top. Under oversupply (very cheap or negative prices), MPVAR plants have a real incentive to CURTAIL - bidding high enough to get skipped rather than accept a near-zero or negative market price.
- * FIT (feed-in tariff) - the plant gets a fixed, guaranteed price regardless of the market price. It has no incentive to curtail under oversupply - it just runs and gets paid its fixed rate.

Found by inspecting a real midday hour agent-by-agent (Phase 13):

MPVAR-subsidised renewables curtailed under the midday glut, FIT-subsidised ones did not - meaning FIT capacity kept adding to the oversupply exactly when it was already worst. (In this build: solar rooftop and run-of-river use FIT; wind onshore/offshore and solar openfield use MPVAR - matching Germany's real subsidy split.)

4. The Public Holiday Check

You asked: "Checked public holidays" - what did we do, in terms of code, analysis, etc?"

No separate saved build - this was a statistical check against the EXISTING price comparison, not a new simulation. Used Python's 'holidays' reference library to get Germany's 5 real nationwide 2027 public holidays that fall on a weekday (New Year's Day, Good Friday, Easter Monday, Ascension Day, Whit Monday), then compared AMIRIS-vs-Brainpool bias on those specific days against ordinary days.

First look was promising, but two follow-up checks killed it:

- (1) Controlling for month - all 5 holidays fall in Jan/Mar/May, months that already have smaller bias regardless of holidays, and New Year's Day actually looked WORSE than an ordinary January weekday once controlled.
- (2) Checking whether any holiday's bias sat meaningfully outside the normal day-to-day spread within its own month - none did. Genuine null result, not pursued further.

5. How 2016 Was Identified as the Closest Weather Match to 2009

You asked: "What did we do to know that 2016 weather data was the closest to 2009, how did we get that?"

Pulled real hourly temperature (DWD, four German weather stations: north/east/middle/southwest) for 2009 and every candidate year with real German demand data available (2015-2019, 2023). Scored each candidate against 2009 on four measures: annual mean temperature, heating-degree-days, monthly-pattern RMSE, and hour-to-hour correlation.

Result:

2023 (the year in use until then) was the WORST match of all six (monthly RMSE 3.37 C, HDD deficit -508). 2016 was the best (RMSE 2.31 C, HDD deficit -40). Full ranking: 2016 < 2018 < 2017 < 2015 < 2019 < 2023 - notably, 2019 (which had been suggested as an alternative) was actually the second-worst.

6. The Correlation Types Used Throughout the Project

You asked: "Define all the correlation types we have been considering, all hours, exclusive shortage, etc."

- * All-hours correlation - every one of the 8,760 hours included. Problem: a handful of extreme shortage-price hours (near 3,000 EUR/MWh) can swing this a lot even when ordinary-hour tracking barely changed - a repeatedly-confirmed outlier-leverage artifact.
- * Excl-shortage correlation - the same calculation with hours at the 3,000 EUR/MWh ceiling removed first. This became the project's trusted metric specifically because it is not distorted by those outliers.
- * Weekday vs. weekend correlation - a split used to diagnose the calendar-alignment bug (Phase 10-11).
- * Hour-to-hour correlation on a specific input series - e.g. the renewables.ninja cross-checks ($r=0.90$ onshore wind, $r=0.97$ solar, $r=0.01$ offshore wind) - a different use of the same statistic, checking two INPUT profiles against each other rather than AMIRIS's output against Brainpool's.
- * "Match quality" in the supervisor deck is just plain language for excl-shortage correlation.

7. Building Price-Responsive Electrolysis and E-Mobility

You asked: "How did we build the "Built price-responsive electrolysis" and "Built price-responsive e-mobility (smart charging)"?"

Both use the exact same real, proven pattern - a GenericFlexibilityTrader modelled as a one-way flexible consumer (the same pattern AMIRIS's own SectorCoupling demo uses for EVs/heat pumps):

- * NetDischargingPowerInMW: 0 - never sells back to the grid, pure consumption.
- * NetInflowPowerInMW - a constant negative drain representing the mandatory annual energy draw (electrolysis: -1,708.45 MW average = 14.966 TWh/yr; e-mobility: -2,032.78 MW average = 17.807 TWh/yr) - continuously depleting an internal energy buffer.
- * An internal buffer (EnergyContentUpperLimitInMWh) that must be periodically recharged - electrolysis: 100,000 MWh (~2.4 days of slack); e-mobility: 40,000 MWh (~0.8 days - deliberately smaller, since EVs have real drivers who need their cars, unlike an industrial electrolyser).
- * Assessment: MAX_PROFIT - the agent CHOOSES WHEN to recharge within its buffer's slack, preferring cheap/negative-price hours over a flat draw.

Verified directly:

Electrolysis hit 14.94 of 14.97 TWh target, peaking 09:00-13:00 (the exact midday surplus window); e-mobility hit 17.80 of 17.81 TWh, flipping from the old evening-peak 'unmanaged charging' shape to a 09:00-13:00 peak / 17:00-19:00 trough - genuine behavioural change, not just a minor shift. E-mobility alone eliminated shortage hours completely for the first time in the project.

8. Recap: The Seasonal Gap Investigation (Phase 23)

You asked: "Give a recap of the seasonal gaps we investigated and how we investigated it."

Re-ran the month/hour/weekday breakdown after the demand-side fixes were in place, to see where the REMAINING gap concentrated. Found three things: (1) the midday gap was still dominant (-41 EUR/MWh at 11:00) and barely moved despite ~33 TWh/year of new flexible demand - there is simply more midday surplus than that much shiftable demand can absorb; (2) weekday/weekend gap had shrunk to ~21 EUR/MWh but was not fully closed; (3) a NEW finding - a July-October seasonal bias (-17 to -23 EUR/MWh vs. -5 to -12 the rest of the year).

For the seasonal gap, five specific causes were checked directly and ruled out one by one:

Renewable output totals (stable, no anomaly), solar/wind mix (similar in both small-gap and large-gap months), import price level (actually cheap in Jul-Oct, not high), base demand levels (similar to May-June, which has a small gap), conventional plant availability (flat all year in this model). What the data DID show: Brainpool's real price climbs smoothly from May to September while AMIRIS's stays flat - traced to the same structural cause flagged earlier: Brainpool's real methodology couples ~30 countries' reservoir drawdown, outages, and gas-storage dynamics; a single-zone Germany model has no mechanism to reproduce that.

9. Cycling/Start-Up Cost, Explained Simply

You asked: ""Tested giving conventional plants a real start-up/cycling cost" - what is cycling/start-up cost? Explain the headline result in a simpler term."

What it is

The real cost a power plant incurs turning off and back on - thermal stress, fuel for reheating, wear. A plant with a genuine start-up cost has a real incentive to keep running at a small loss (even bidding slightly negative) rather than shut down and pay to restart later.

Real numbers sourced

A peer-reviewed study (Roques/Hach et al., Nature Energy 2017) gave real figures - roughly EUR50,000-70,000 per start for a large lignite/coal block, EUR60,000 for gas - converted into AMIRIS's units (62.5/87.5/120 EUR/MW).

Simple result:

Built and ran the actual test. The output came back BYTE-IDENTICAL to the baseline - every one of the 8,760 hours, to the cent. Why: AMIRIS's own code genuinely calculates this cost internally, but that calculation is never actually plugged into the number a power plant offers when it bids into the market. Like flipping a light switch that was never wired to anything - the switch is real, the calculation happens, but nothing downstream ever uses it.

10. Data Gap Sources, and the Offshore Wind Cross-Check

You asked: "Provide the sources for the external data used to resolve data gaps, and how did we verify AMIRIS's wind offshore profile with renewables.ninja?"

Real external sources used throughout: EEG (Erneuerbare-Energien-Gesetz, Germany's renewable law) and BNetzA (Bundesnetzagentur) auction data for subsidy rates; MaStR (Marktstammdatenregister) for battery storage duration; DWD (German weather service) for real temperature; renewables.ninja for real 2009 wind/solar weather; ENTSO-E and Eurostat for cross-border data (later, the market-coupling build).

Offshore wind verification, specifically

Brainpool never supplied its own offshore shape at all - AMIRIS's own example wind_offshore_profile.csv was reused as a placeholder from the very start. The 'verification' was comparing that reused placeholder against a genuine, independent real-2009-weather offshore profile pulled from renewables.ninja.

Technology	Has a real Brainpool shape?	Correlation vs. real 2009 weather
Wind onshore	Yes	r = 0.896
Solar (openfield)	Yes	r = 0.968
Wind offshore	No - AMIRIS's own placeholder reused	r = 0.012

Honest finding:

Offshore wind's correlation is essentially zero, even though the average capacity factor was in a similar ballpark (0.371 vs. 0.469). Offshore wind's shape is the one genuinely unverified assumption in the whole renewable build - flagged as such rather than glossed over.

11. Where the Import Ceiling Is Actually Set

You asked: "The import ceiling tweaks - how were we implementing it? Are we setting the limit in the schema or scenario file, or just using a timeseries file?"

Confirmed directly from the agent file (Import.yaml):

It is a plain ImportTrader agent whose AvailableEnergyForImport attribute points to a TIMESERIES CSV FILE, not a scalar value in the schema or scenario file. AMIRIS's schema does not offer a 'constant ceiling' attribute option - even a flat ceiling has to be written out as the same number repeated for all 8,760 hours (confirmed: AvailableEnergyForImport.csv literally contains '30000.0000' on every line). ImportCostInEURperMWh.csv is the same pattern, holding the real French day-ahead price series.

12. The Lignite Bidding Formula

You asked: "What is the lignite bidding formula?"

Decompiled directly from ConventionalTrader.prepareBids:

For each generation block: bid price = marginal cost + a linearly-interpolated markup, where the markup runs from minMarkup (applied to the cheapest block in the fleet) to maxMarkup (applied to the most expensive block). Lignite's minMarkup was -60 EUR/MWh - the widest negative-bidding allowance of any fuel (gas: -10, hard coal: -15) - confirmed as a genuine, working mechanism (unlike the cycling cost).

One caveat found in the same decompilation:

Lignite's own marginal cost means even the full -60 markup can only push the bid to roughly -15 to -25 EUR/MWh - the deeper troughs actually observed (-66 to -85) come from a DIFFERENT mechanism: blocks tagged MustRunFactor bid at a flat -500 EUR/MWh regardless of markup.

13. Data Sourced for the Market-Coupling Export Build

You asked: "What data did we source for the market coupling export build?"

* Real Eurostat data (nrg_cb_e for demand, nrg_inf_epc for generation capacity by technology) for 9 of Germany's 10 real neighbours.

- * Switzerland backfilled via ENTSO-E (Eurostat has no non-EU coverage).
- * renewables.ninja for real 2009-weather wind/solar profiles per country, capacity-weighted.
- * ENTSO-E cross-border physical flow data (after working through a real multi-day platform outage) to derive real transmission capacity.
- * IHA 2023's real EU-wide pumped-storage share (30%) and Switzerland's real reservoir:run-of-river ratio (90.2:9.8) for the hydro storage split.
- * Germany's own real BNetzA-auction subsidy levels, reused as the best available reference point (no separate per-country subsidy scheme was researched).

14. Every Export/Market-Coupling Variant Actually Tried

You asked: "Explain clearly all the various things/models we tried with export - placeholder, combining 10 countries, building each country individually, etc."

Phase 34 - Placeholder transmission

First working two-zone build, a flat 30,000 MW guess (reused from the old import ceiling) for the border capacity.

Phase 36 - Real-flow transmission

Replaced the guess with a real number derived from actual 2023 ENTSO-E flow data (98th percentile of hourly flow, ~21,600-22,000 MW).

Phase 37 - Real storage + subsidy (BEST RESULT)

Gave the combined 'Rest of Europe' zone real battery/reservoir storage and real subsidy treatment instead of a placeholder zero.

Phase 42 - France disaggregated alone

Split France out of the combined zone into its own real, separately-modelled zone. Made things WORSE (shortage hours 7 to 167) - France's nuclear fleet had been propping up the whole combined pool.

Phase 43 - All 10 real countries disaggregated

Went all the way. Better than France-alone but still short of the combined-zone result on the metrics that matter most - so the combined 'Rest of Europe' approach (Phase 37) remains the standing best build.

15. How the Two-Way Market Works: One Import Hour, One Export Hour (Real Data, Phase 37)

You asked: "How does the 2-way market work with the export? Set up an example of an hour that had export and another hour that had import."

	Import hour: 2027-12-31, 18:00	Export hour: 2027-10-19, 11:00
Germany's own price	153.92 EUR/MWh	-32.11 EUR/MWh
Rest-of-Europe price	88.79 EUR/MWh	64.97 EUR/MWh
Germany imported	22,011.7 MWh	0 MWh
Germany exported	0 MWh	21,593.4 MWh

Import hour explained:

Germany's own price was much higher than the Rest-of-Europe price, so Germany imported 22,011.7 MWh - essentially the ENTIRE available transmission capacity that hour (the real ceiling was 22,012 MW) - because it was clearly cheaper to buy from across the border than to run its own more expensive generation.

Export hour explained:

Germany's own price was actually NEGATIVE (a real midday oversupply hour - cheap solar/wind flooding the market) while the Rest-of-Europe price was still positive, so Germany exported 21,593.4 MWh - again essentially the full available capacity in that direction (21,593 MW) - because it was more profitable to sell the surplus abroad than to let it go to waste or push the domestic price even more negative.

Across the full year:

4,920 hours had some import, 3,752 had some export - the model genuinely trades in both directions depending on which side is cheaper that hour, which is the entire point of the two-way coupling versus the old one-way import-only model.